

Capacitance & Capacitors & Dielectrics

Siġa, siġar, kondansatör

$$C \equiv \frac{Q}{\Delta V} = \text{positive}$$

capacitance

$$Q = +$$

$$\Delta V = +$$

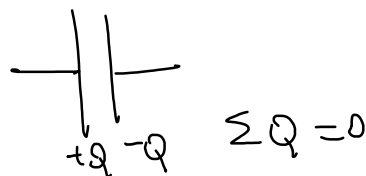
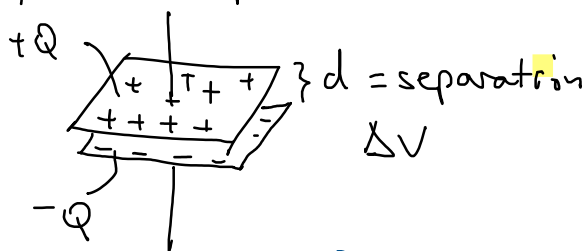
$$\frac{q}{\Delta V}$$

$$\rightarrow \text{Farad} = \left[F = \frac{C}{V} \right]$$

1F is a BIG number.

$$\frac{Q \sim 1\text{nc} \quad 1\mu\text{c} \quad 1\text{mC}}{\Delta V \sim 1\text{V} \quad \dots \quad 100\text{V}} = \text{nf},$$

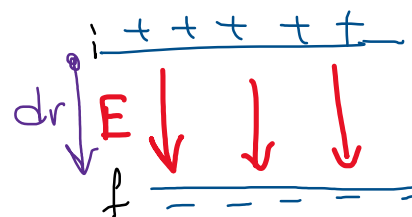
parallel plates



$$C = \frac{Q}{\Delta V}$$

$$\Delta V = - \int \vec{E} \cdot d\vec{r}$$

uniform constant.



$$\Delta V = - \int E dr \cos 0$$

$$= -E \int dr = -Ed = V_f - V_i$$

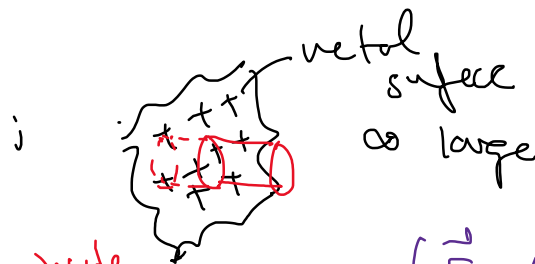
separation = d

$$\Delta V < 0$$

$$\Delta V = \text{absolute}$$

$$\Delta V = Ed$$

$$C = \frac{Q}{Ed}$$



inside
 $E=0$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

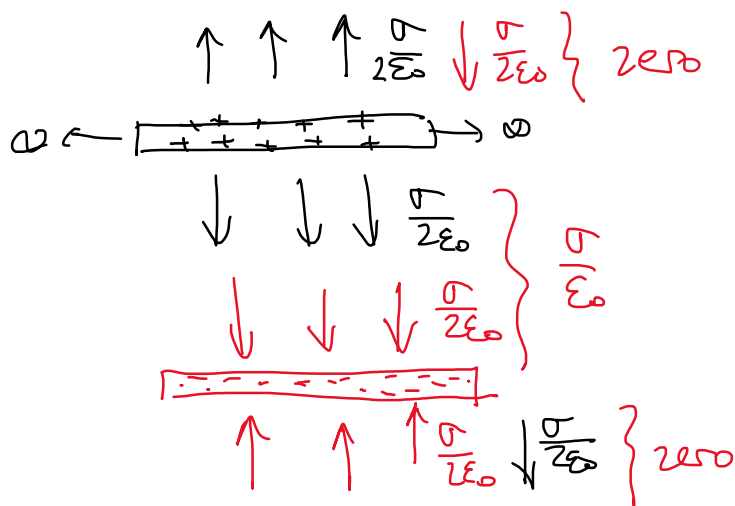
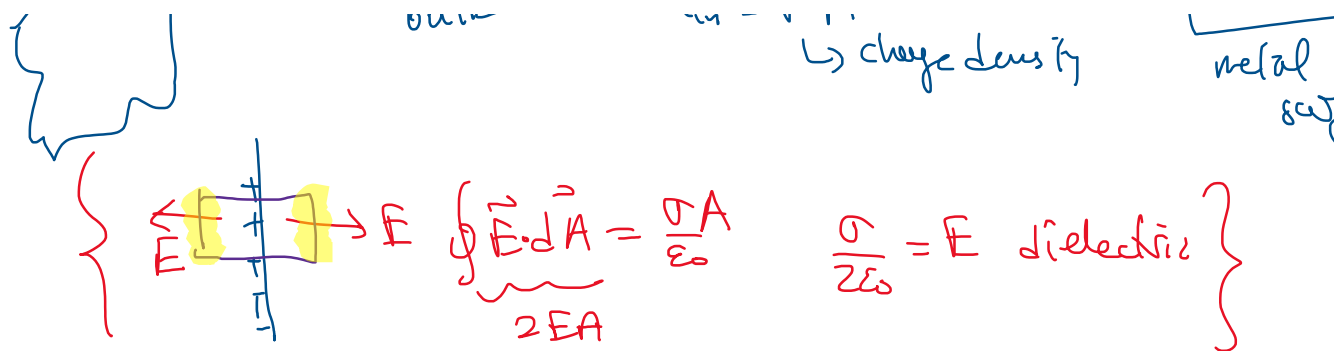
$$EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

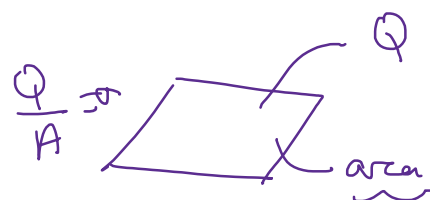
dielectric
 ∞ surface

metal
bulk

$$q = \sigma A$$



$$C = \frac{Q}{\Delta V} = \frac{Q}{Ed} = \frac{Q}{\frac{\sigma}{\epsilon_0} d} = \frac{\sigma A \epsilon_0}{\sigma d}$$



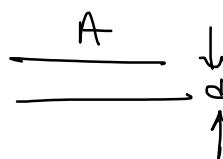
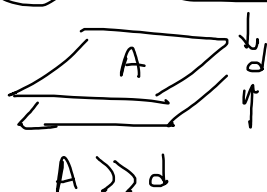
$$C = \epsilon_0 (\text{Length})$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad \epsilon_0 = \left[\frac{C^2}{Nm^2} \right]$$

$$F = \left[\frac{C}{V} \right] = \left[\frac{C^2}{Nm^2} \right]$$

$$C = \epsilon_0 \frac{A}{d}$$

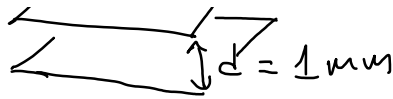
parallel



$d \downarrow C \uparrow$
 $A \uparrow C \uparrow$

Akdeniz university campus
 Area $\sim 3 \times 10^6 \text{ m}^2$

$$C = 8.85 \times 10^{-12} \frac{3 \times 10^6 \text{ m}^2}{10^{-3} \text{ m}}$$



$$\approx 25 \times 10^{-3} \text{ F} \approx 25 \text{ mF}$$

$$\text{Area} = 3 \times 40 \times 10^6 \text{ m}^2 \quad C = 1 \text{ f}$$

ex) parallel plate $2 \text{ m}^2 = A$; $d = 5 \text{ mm}$

$$\Delta V = 10 \text{ kV} \quad Q = ?$$

$$C = \frac{Q}{\Delta V} = \epsilon_0 \frac{A}{d} = 8.85 \times 10^{-12} \frac{2 \text{ m}^2}{5 \times 10^{-3}} = 3.5 \text{ nF}$$

$$Q = C \Delta V = 3.5 \times 10^{-9} \times 10 \times 10^3$$

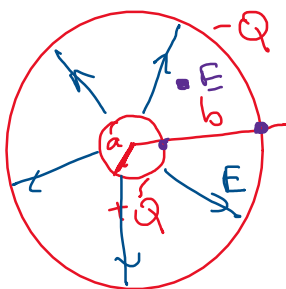
$$= 3.5 \times 10^{-5} = \underline{\underline{35 \mu\text{C}}}$$

Electric field between plates?

$$E d = \Delta V$$

$$E = \frac{10000}{5 \times 10^{-3}} = 2 \times 10^6 \text{ V/m}$$

spherical capacitors



$$C = ?$$

$$C = \frac{Q}{\Delta V}$$

netel $\oint \vec{E} \cdot d\vec{r} = \frac{Q}{\epsilon_0}$

$$\Delta V = - \int \vec{E} \cdot d\vec{r}$$

b = outer radius

$$= - \frac{Q}{4\pi\epsilon_0} \int \frac{dr}{r^2} = + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)$$

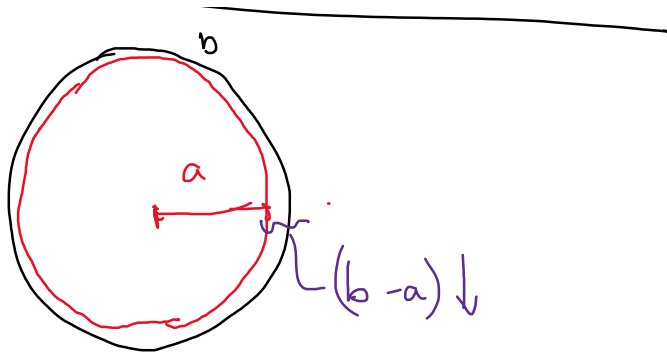
a = inner radius

$$\Delta V = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)$$

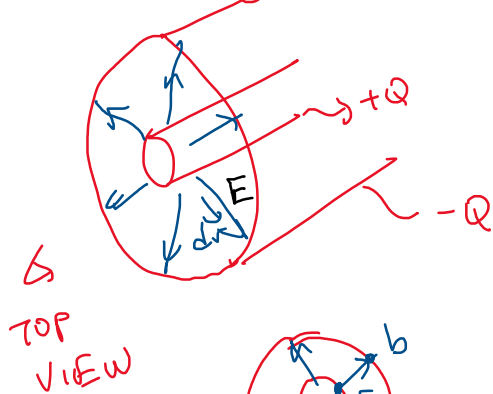
$$C = \frac{Q}{\Delta V} = \frac{Q}{\frac{Q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)}$$

$$C = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right) = \epsilon_0 (\text{Length})$$

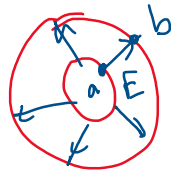
$C \uparrow$ $(b-a) \downarrow$
 $a, b \uparrow$



cylindrical capacitors



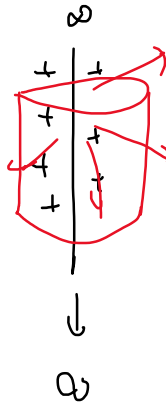
TOP VIEW



$$\frac{Q}{\Delta V} = C$$

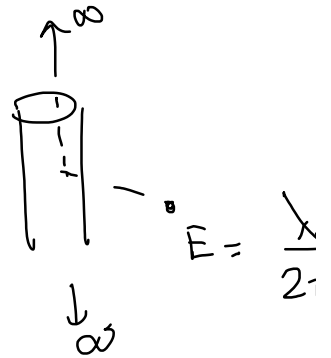
$$\Delta V = - \int \vec{E} \cdot d\vec{r}$$

Gauss Law
Box



$$E = \frac{\lambda}{2\pi\epsilon_0 r} \rightarrow \frac{Q}{L}$$

a = inner rad.
b = outer rad.



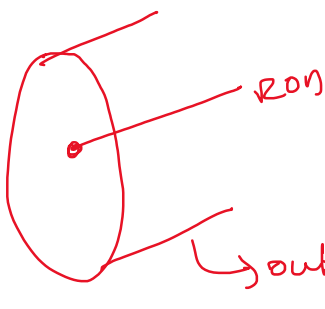
$$\Delta V = - \int \vec{E} \cdot d\vec{r} = - \int_a^b \frac{\lambda}{2\pi\epsilon_0} \frac{dr}{r}$$

$$= - \frac{\lambda}{2\pi\epsilon_0} \int_a^b \frac{dr}{r} = - \frac{\lambda}{2\pi\epsilon_0} [\ln r]_a^b = - \frac{\lambda}{2\pi\epsilon_0} (\ln b - \ln a)$$

$$\Delta V = - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}$$

$$C = \frac{Q}{\Delta V} = \frac{Q = \lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{b}{a} \right)}$$

$$Q \leftrightarrow \lambda$$

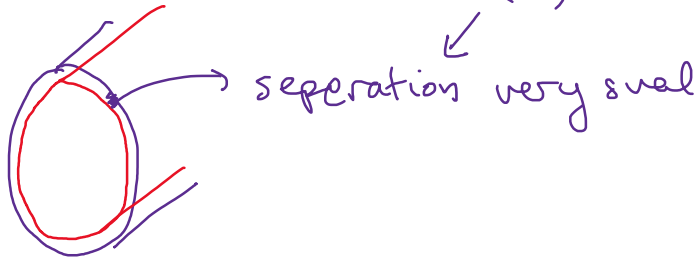


$$\lambda L = Q$$

$$2\pi r A = Q$$

$$C = \frac{\lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}} = 2\pi\epsilon_0 \frac{L}{\ln(\frac{b}{a})} \Rightarrow \epsilon_0 \cdot \ln \frac{b}{a} =$$

$$C \uparrow ; L \uparrow ; \ln(\frac{b}{a}) \downarrow \quad b \geq a$$



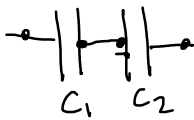
$$C = \frac{Q}{\Delta V} \uparrow$$



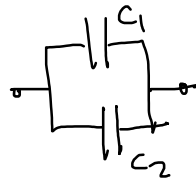
$$C = \epsilon_0 (\text{Length}) \approx \frac{nF}{\frac{nF}{mF}} \quad \text{geometry}$$

capacitors in parallel & series connections

series



parallel

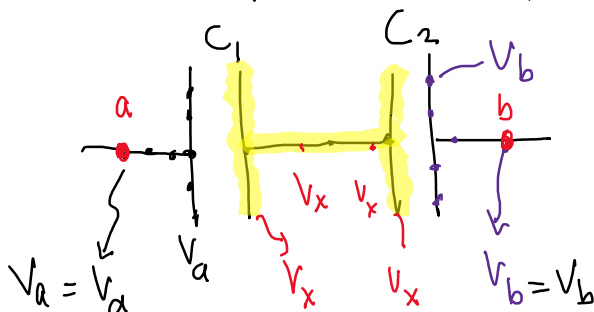


important reminder
metal surfaces are

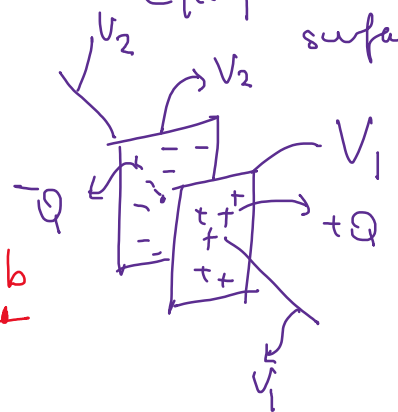
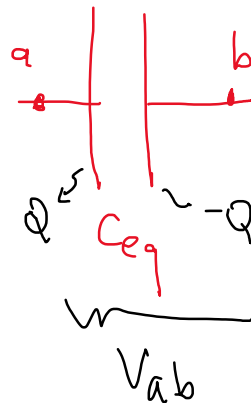
equipotential surface



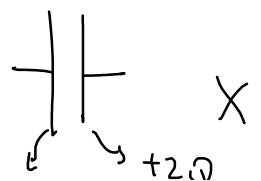
C_{eq} = equivalent capacity



$\equiv$



$$C = \frac{Q}{\Delta V}$$



$$V_{ab} = V_{ax} + V_{xb}$$

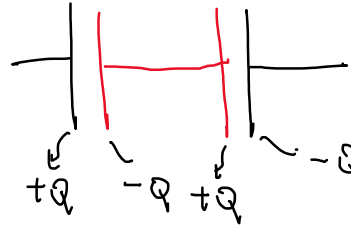
$$C_1 = \frac{Q_1}{V_{ax}} = \frac{Q}{V_{ax}}$$

$$C_2 = \frac{Q_2}{V_{bx}} = \frac{Q}{V_{bx}}$$

$$\psi_1 = \psi_2$$

$$Q_1 = Q_2 = Q$$

$$Q = V_{ax} C_1 = V_{bx} C_2 = V_{ab} C_{eq}$$



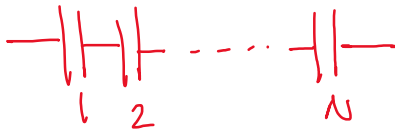
$$C_{eq} = \frac{Q}{V_{ab}}$$

$$\frac{Q}{C_{eq}} = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

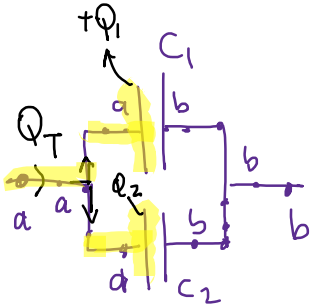
series

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

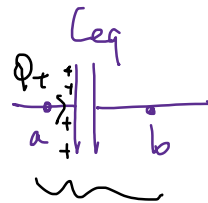


$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

parallel



$\Rightarrow$



$$C_{eq} = \frac{Q_T}{V_{ab}}$$

$$V_1 = V_{ab} = V_2 = V_{ab}$$

$$C_1 = \frac{Q_1}{V_{ab}}$$

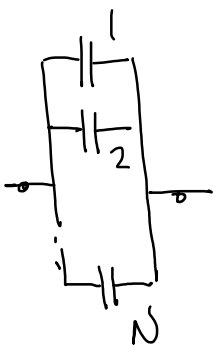
$$C_2 = \frac{Q_2}{V_{ab}}$$

$$Q_1 \neq Q_2$$

$$Q_1 + Q_2 = Q_T$$

$$V_{ab} C_1 + V_{ab} C_2 = V_{ab} C_{eq}$$

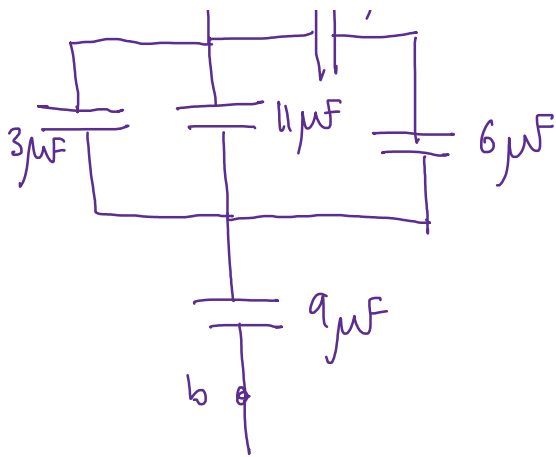
$$C_1 + C_2 = C_{eq} \quad \text{parallel}$$



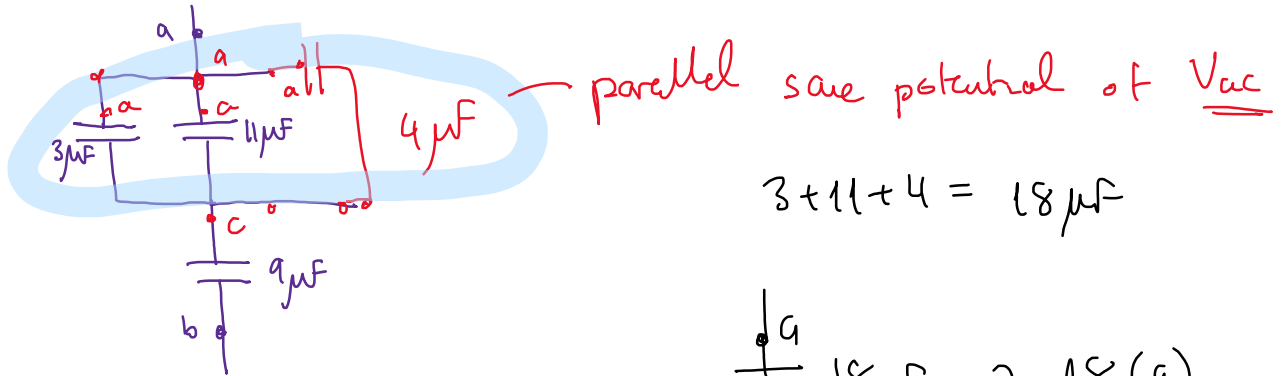
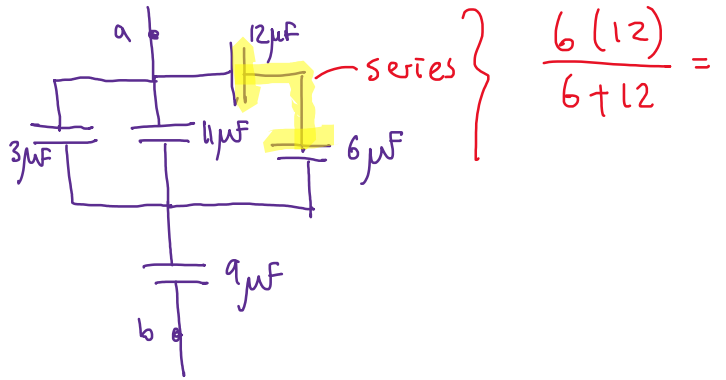
$$C_{eq} = C_1 + C_2 + \dots + C_N$$

a b 11 12 μF

C_{ab} = equivalent capacity



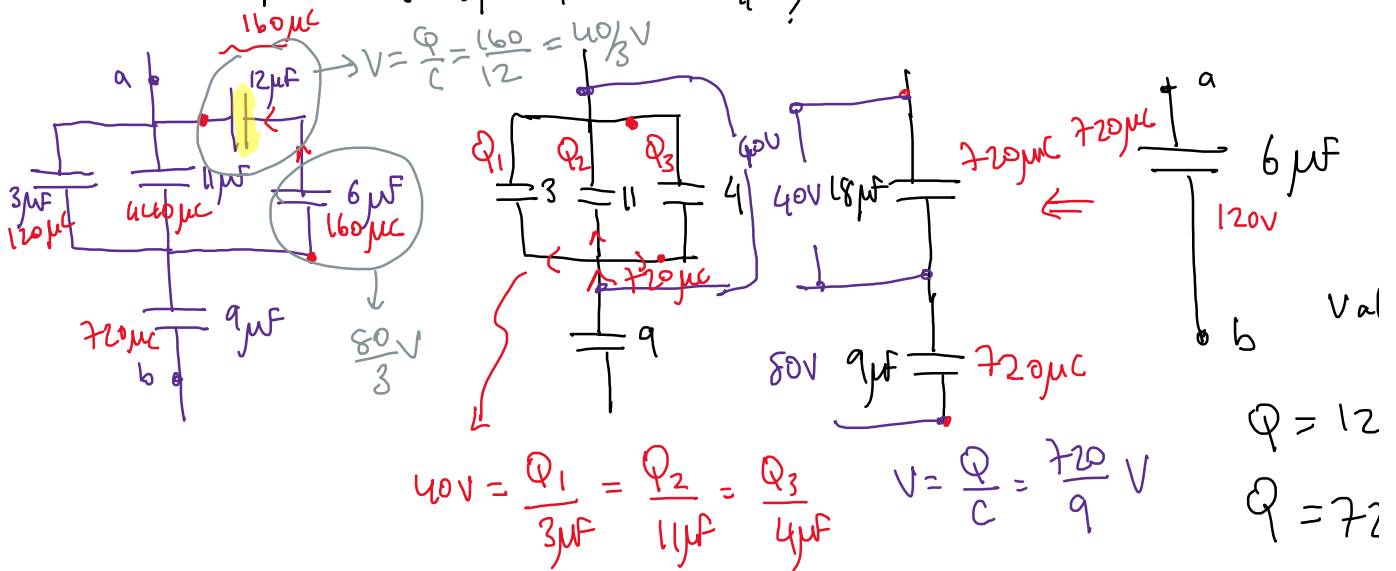
between a & b points



If $V_{ab} = 120V$

find charges on each capacitor?

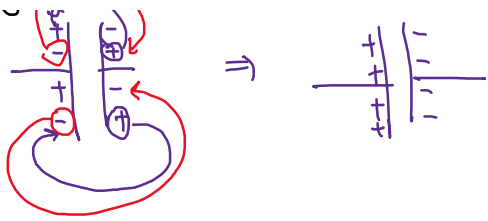
potentials of " " ?



$$\left\{ \begin{array}{l} Q_1 = 120\mu C \\ Q_2 = 40\mu C \\ Q_3 = 160\mu C \end{array} \right\} 720\mu C$$

Energy stored in capacitors.

1 = , 1 = ,



$$W = \int \mathbf{F} \cdot d\mathbf{r} = \int q\mathbf{E} \cdot d\mathbf{r} = -\Delta U$$

$$W = qV$$

$$\Delta U = q \Delta V$$

$$U = \int q \frac{dq}{C} = \frac{q^2}{2C} = \frac{CV^2}{2} = \frac{qV}{2}$$

$$q = CV \quad C = \frac{q}{V}$$

$$\leftarrow \int dU = \int q dV$$

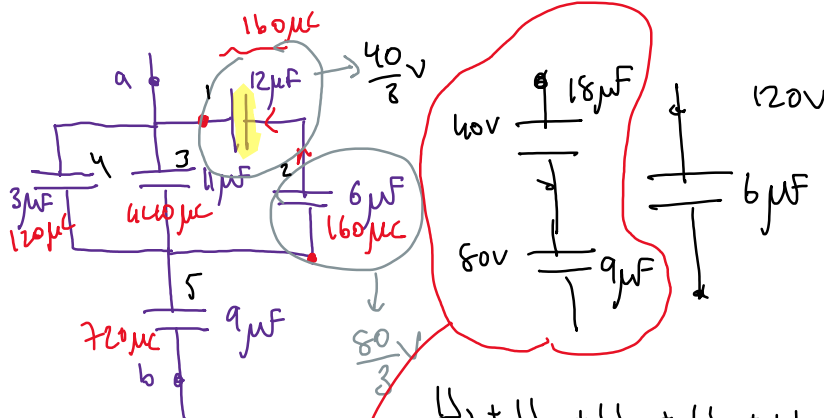
$$V \leftrightarrow q \text{ re}$$

$$C = \frac{q}{V}$$

$$V = \frac{q}{C}$$

$$dV = \frac{dq}{C}$$

$$U = \left[\text{Joules} \right] = \left[\frac{C^2}{F} \right] = [FV^2] = [CV]$$



$$U = \frac{CV^2}{2} = \frac{6(120)^2}{2}$$

$$= 43200 \mu J$$

$$= 43.2 mJ$$

$$U_1 + U_2 + U_3 + U_4 + U_5 = 43.2 mJ$$

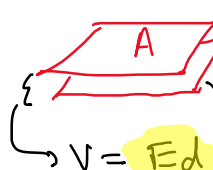
$$U_1 + U_2 = 43.2 mJ$$

$$18 \frac{40^2}{2} + 9 \frac{80^2}{2} =$$

$$14400 + 28800 = 43200 \mu J$$

Energy density of a capacitor

$$u = \frac{\text{energy}}{\text{volume}}$$



$$V = Ed$$

$$E = \frac{\sigma}{\epsilon_0}$$

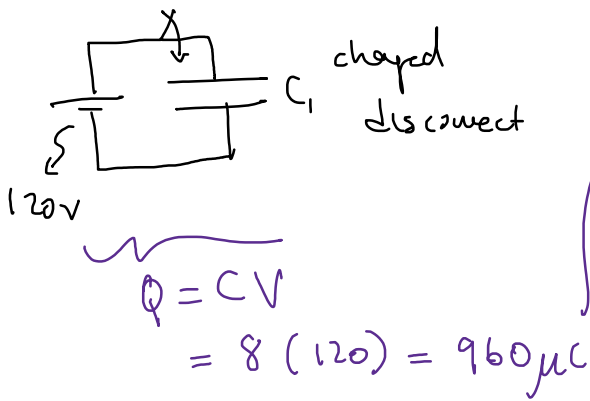
$$u = \frac{CV^2/2}{Ad = \text{volume}} = \frac{C E^2 d^2}{2 Ad} = \frac{C \sigma^2}{2 A \epsilon}$$

$$C = \epsilon \frac{A}{d}$$

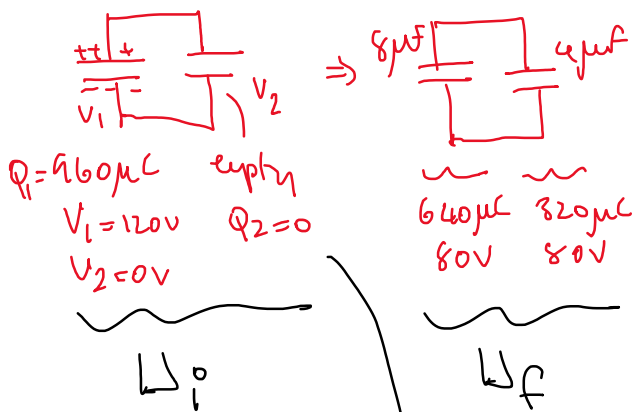
$$u = \frac{\epsilon_0 A \sigma^2 d}{2 A d} = \frac{\sigma^2}{2}$$

$$d = 24 \epsilon_0^2 \quad \underline{\underline{2 \epsilon_0}}$$

(ex) $C_1 = 8 \mu F$; has a charge of $960 \mu C$. (charged with $120 V$)
 C_1 is disconnected from battery and connected to $C_2 = 4 \mu F$ as shown. find charges and potentials for each capacitor.



summary

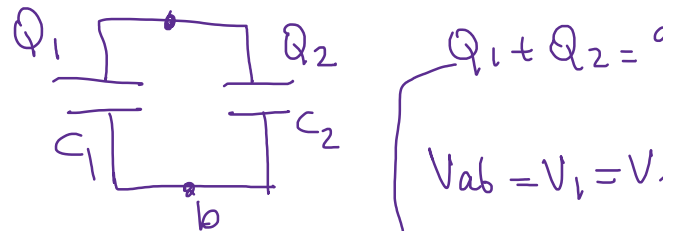
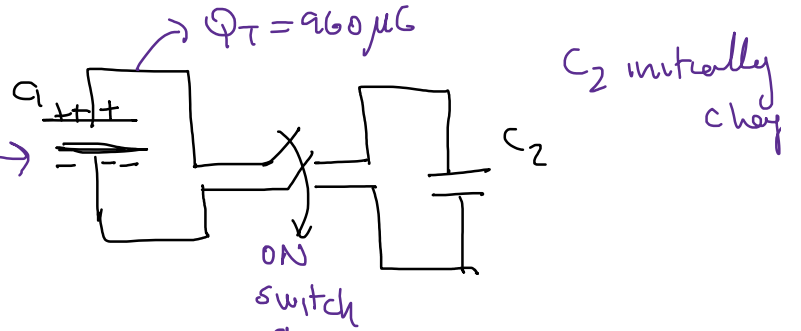
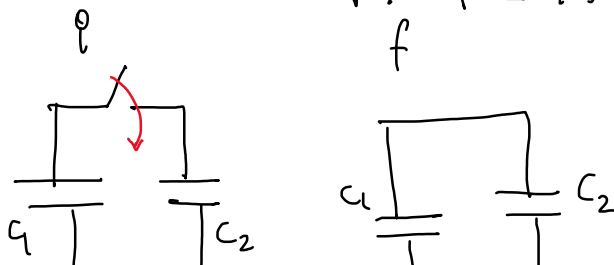


$$U = \frac{1}{2} CV^2$$

$$U_i = \frac{1}{2} 8 (120)^2 + 0 = 57600 \mu J$$

$$U_f = \frac{1}{2} 8 (80)^2 + \frac{1}{2} 4 (80)^2 = 38400 J = \frac{1}{2} C_{eq} V_{ab}^2 = \frac{1}{2} 12 (80)^2$$

$$U_i > U_f = ??$$



$$V_{ab} = \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$Q_1 = C_1 V_{ab} = 8 \mu F 80V$$

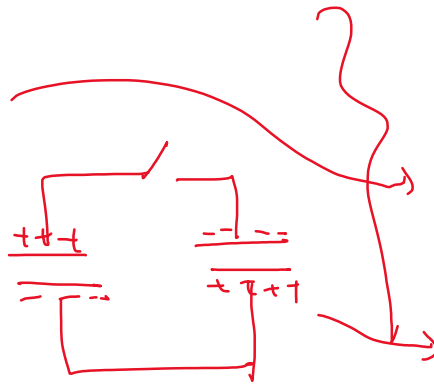
$$8 V_{ab} + 4 V_{ab} = 96$$

$$V_{ab} = 80V$$

$$960 \mu C \begin{cases} Q_1 = 640 \mu C \\ Q_2 = 320 \mu C \end{cases}$$



- i) C_2 is also charged.
- ii) C_2 is charged but the polarity is opposite

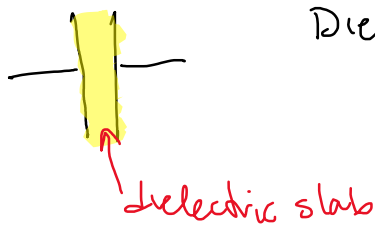


$Q_T = 15 \text{ } \mu\text{C}$ is conserved

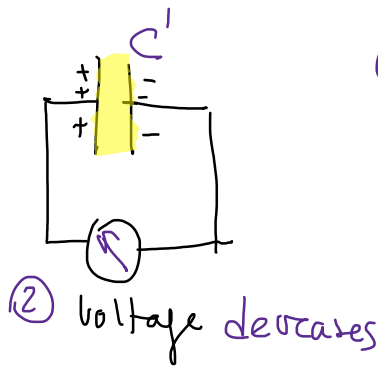
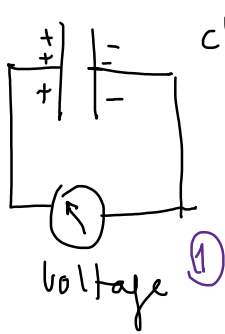
$$\left. \begin{aligned} Q_{1i} &= 100 \mu\text{C} \\ Q_{2i} &= 50 \mu\text{C} \end{aligned} \right\}$$

$$\left. \begin{aligned} Q_{1f} &= 100 \mu\text{C} \\ Q_{2f} &= -50 \mu\text{C} \end{aligned} \right\}$$

Dielectrics (Insulator)



Dielectrics are used to increase the capacity of a capacitor
it also increases the stability of cap



$$C = \frac{Q}{\Delta V} \quad C' = \frac{Q'}{\Delta V'}$$

$$Q' = Q \text{ same}$$

$$C' \uparrow$$

$$C \Delta V = C' \Delta V'$$

$$C' > C$$

$\kappa = \text{Kappa} = \text{dielectric const}$

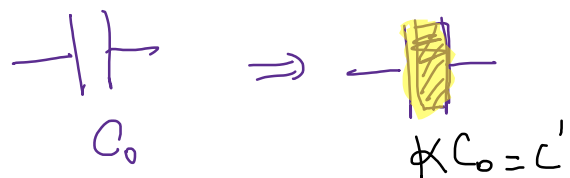
① C
 ΔV

② $C' = \kappa C$

$$\Delta V' = \frac{\Delta V}{\kappa} \Rightarrow \frac{E_0}{\kappa} = E' \quad \kappa \geq 1$$

$\hookrightarrow$ empty space $\kappa = 1$

κ	natural
1.0000000000	vacuum
1.00006	air
2.1	teflon
...	...



5-10

42.5

310

glass

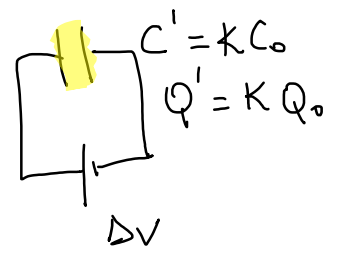
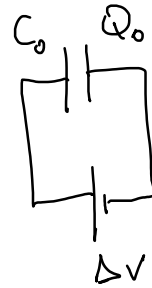
glycerin

strontium
titanate

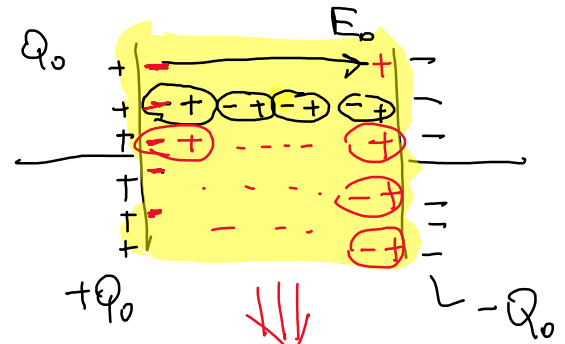
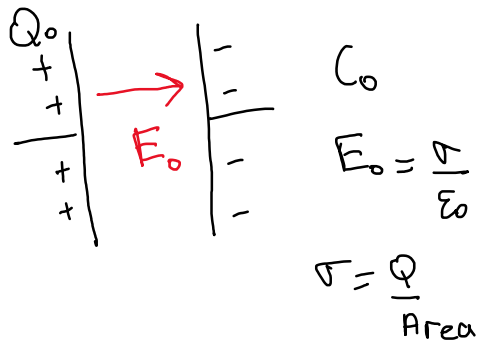
1 nF

→ 4 nF

$K=4$



$\vec{E}$ within the capacitor when K dielectric material is inserted.



within the dielectric



$$E' = E_0 - E_i$$

$$E' < E_0 ; \Delta V' < \Delta V$$

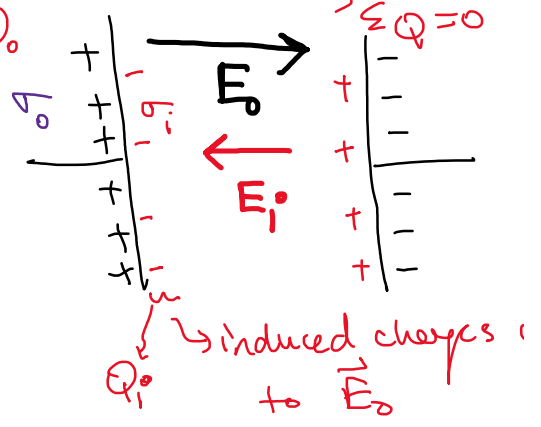
$$\Delta V = E d$$

$$\Delta V' = \frac{\Delta V}{K}$$

$$E' = E_0 - E_i = \frac{E_0}{K}$$

$$\frac{E_0}{K} = \frac{\sigma_0}{K \epsilon_0}$$

E_i = induced field in dielectric material
 $Q_i < Q_0$



$$-\frac{E_0}{K} + E_0 = E_i$$

$$E_0 \left(1 - \frac{1}{K}\right) = E_i = E_0 \left(\frac{K-1}{K}\right)$$

$$E_i = E_0 \frac{K-1}{K} \Rightarrow \frac{\sigma_i}{\epsilon_0} = \frac{\sigma_0}{\epsilon_0} \left(\frac{K-1}{K}\right) \Rightarrow \sigma_i = \sigma_0 \left(\frac{K-1}{K}\right)$$

$$K=2$$

$$Q_0 = Q_0 \cdot 2-1 = Q_0$$

$$Q_i = Q_0 \left(\frac{K-1}{K}\right)$$

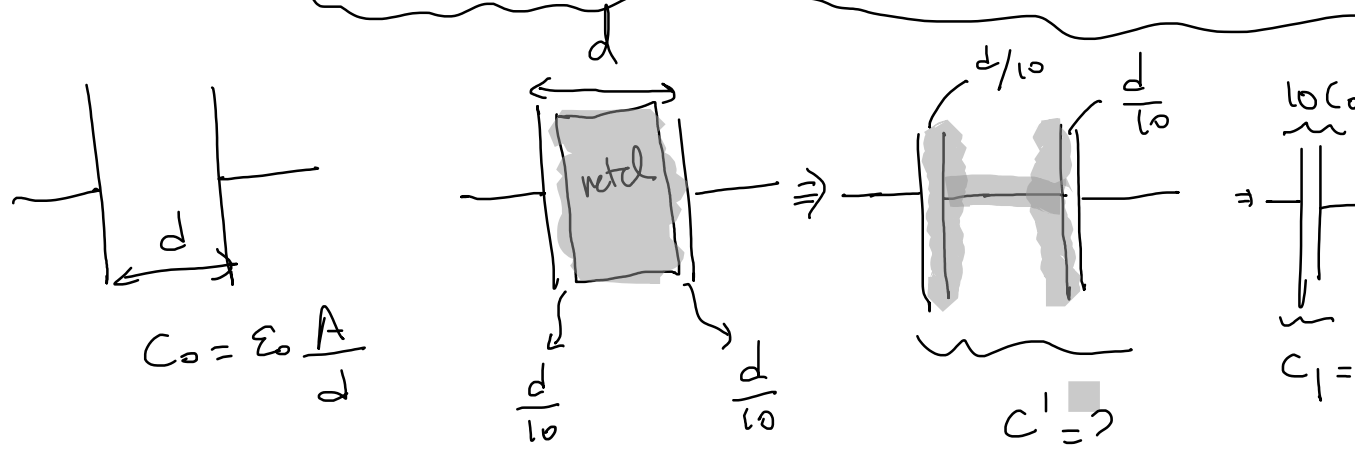
$$\varphi_i - \varphi_0 = \frac{V_0}{2} = \frac{10}{2}$$

$$K = 300$$

$$Q_i^0 = Q_0 \frac{299}{300}$$

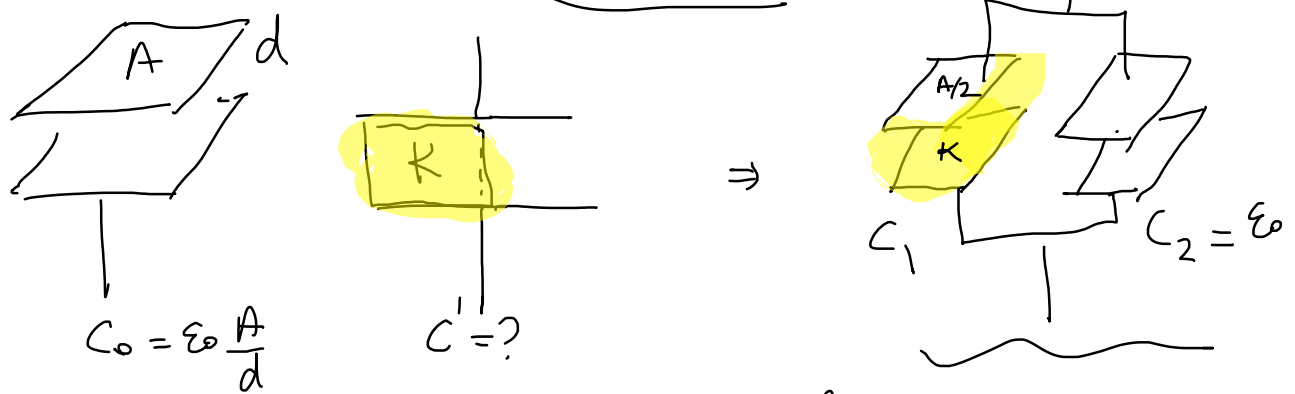
$$\varphi_i - \varphi_0 = \frac{V_0}{2}$$

$$\varphi_i^0 = \varphi_0 \quad K \rightarrow \infty \Rightarrow \underline{\underline{\text{retol}}}$$



$$\text{series} \Rightarrow \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \frac{C_1 C_2}{C_1 + C_2} = \frac{10C_0 (10C_0)}{20C_0}$$

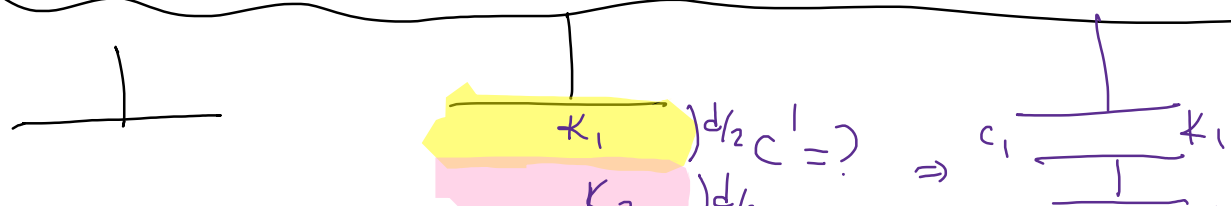
$$C' = 5C_0$$



$$C_1 = \epsilon_0 K \frac{A/2}{d} \quad C' = C_1 + C_2$$

$$C_1 = K \epsilon_0 \frac{A}{2d} = \frac{K}{2} C_0$$

$$C' = C_1 + C_2 = \left(\frac{K}{2} + \frac{1}{2} \right) C_0 = \left(\frac{K+1}{2} \right) C_0$$



$$C_0 = \epsilon_0 \frac{A}{d}$$

$$C_1 = K_1 \epsilon_0 \frac{A}{d/2}$$

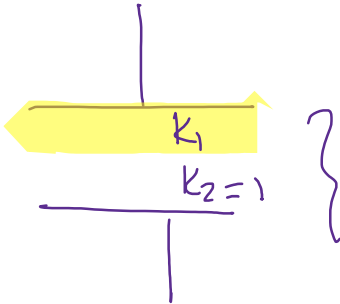
$$C_1 = 2K_1 C_0$$

$$C_2 = 2K_2 C_0$$

$$C_2 = K_2 \frac{A}{d}$$

$$C' = \frac{C_1 C_2}{C_1 + C_2}$$

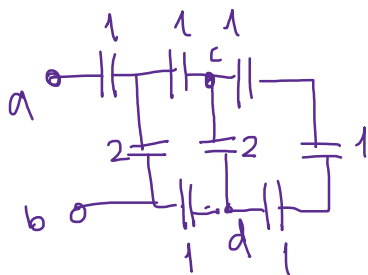
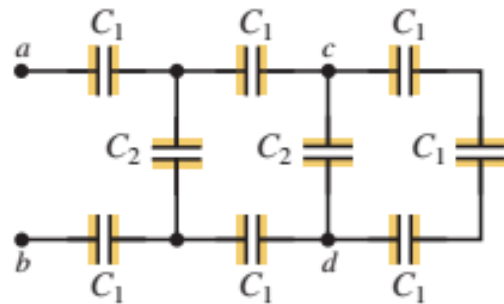
$$C' = \frac{2K_1 K_2 C_0}{2(K_1 + 1)}$$



$$C' = \frac{4K_1 C_0^2}{2C_0(K_1 + 1)} = \frac{2K_1 C_0}{(K_1 + 1)}$$

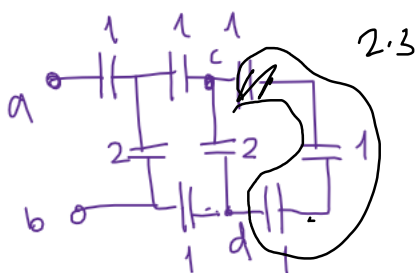
24.63 •• In Fig. P24.63, each capacitance C_1 is $6.9 \mu\text{F}$, and each capacitance C_2 is $4.6 \mu\text{F}$. (a) Compute the equivalent capacitance of the network between points a and b . (b) Compute the charge on each of the three capacitors nearest a and b when $V_{ab} = 420 \text{ V}$. (c) With 420 V across a and b , compute V_{cd} .

Figure **P24.63**



series $\left(\frac{1}{C_1} + \frac{1}{C_1} + \frac{1}{C_1} \right)^{-1} =$

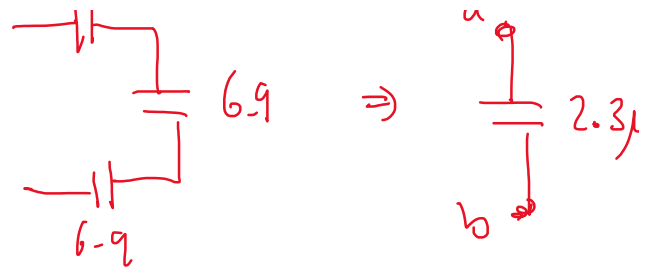
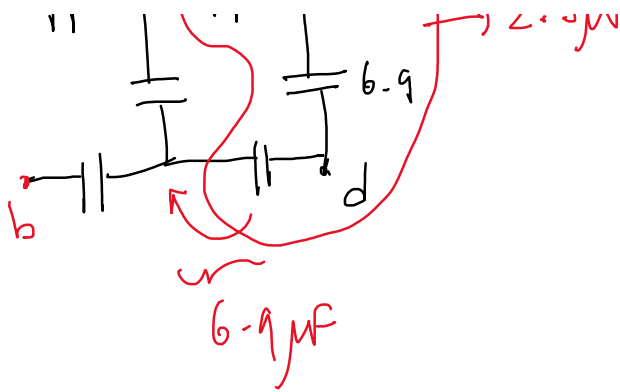
$$\frac{C_1}{3} = \frac{6.9}{3} =$$



$$2.3 + 4.6 = 6.9$$

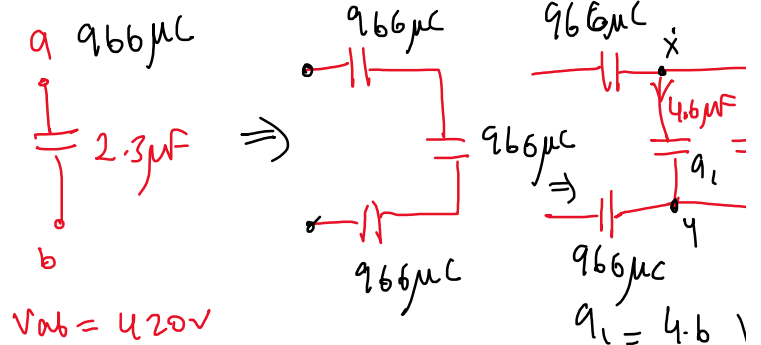
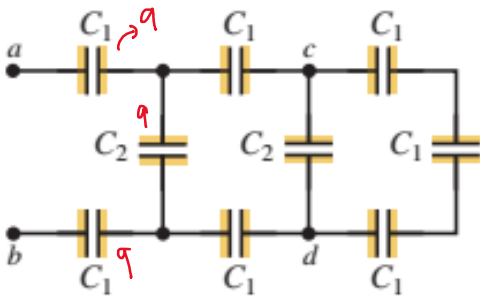


$$6.9$$



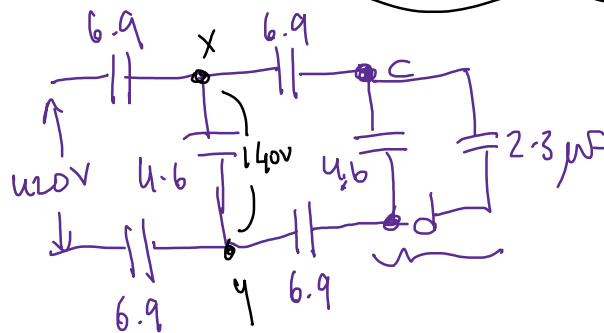
$$Q = C \Delta V$$

$$(2.3 \times 420) \mu C$$



$$V_{ab} = 420V$$

$$V_{cd} = ?$$

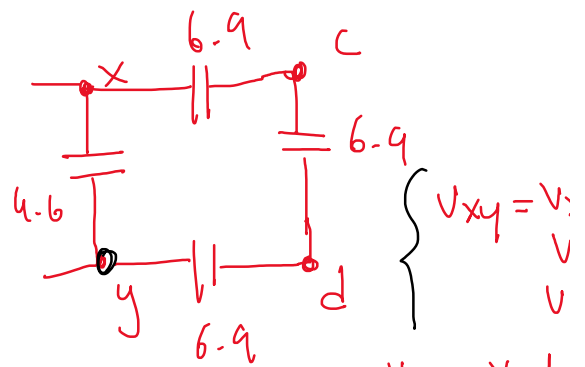


$$V_{xy} \Rightarrow$$

$$q_1 = 4.6 V \times 4$$

$$\frac{644}{4.6} = V_{xy} = 140V$$

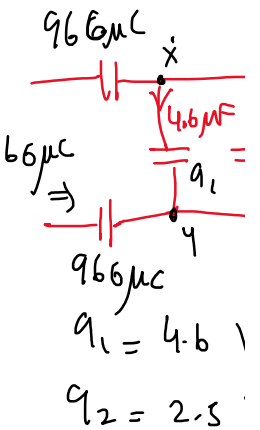
$$V_{xy} =$$



$$V_{xc} = V_{cd}$$

$$V_{xy} = 3 V_{cd}$$

$$V_{cd} = \frac{140}{3} V$$



$$\frac{q_1}{q_2} = 2$$

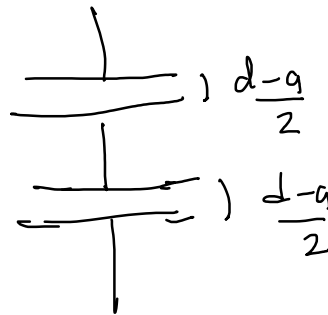
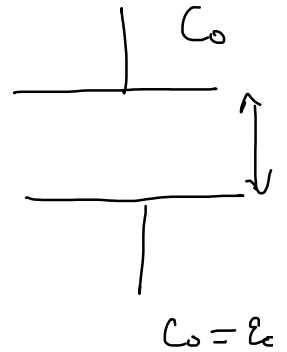
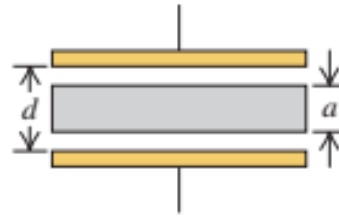
$$966 \mu C = q_1 +$$

$$q_2 = 2$$

$$q_1 = 6$$

24.66 •• An air capacitor is made by using two flat plates, each with area A , separated by a distance d . Then a metal slab having thickness a (less than d) and the same shape and size as the plates is inserted between them, parallel to the plates and not touching either plate (Fig. P24.66). (a) What is the capacitance of this arrangement? (b) Express the capacitance as a multiple of the capacitance C_0 when the metal slab is not present. (c) Discuss what happens to the capacitance in the limits $a \rightarrow 0$ and $a \rightarrow d$.

Figure P24.66



$$C_1 = \epsilon_0 \frac{2A}{(d-a)} = C_2$$

$$\frac{C_1 \cdot C_2}{C_1 + C_2} = \frac{C_1^2}{2C_1} = \frac{C}{2}$$

$$\frac{C_1}{2} = \frac{\epsilon_0 2A}{(d-a)} \frac{1}{2} = \frac{\epsilon_0 A}{d-a}$$

$$a \rightarrow 0$$

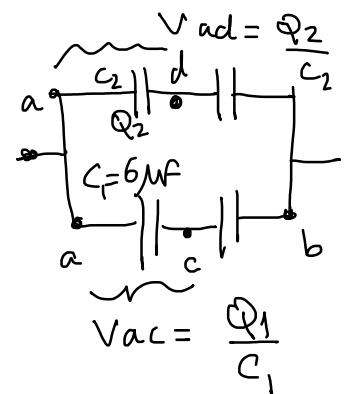
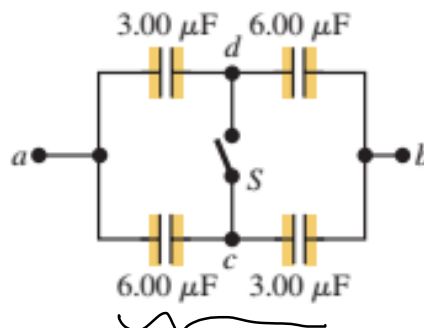
$$C' \rightarrow C_0$$

$$a \rightarrow d$$

$$\frac{\epsilon_0 A}{d-a} \rightarrow \infty$$

24.60 • The capacitors in Fig. P24.60 are initially uncharged and are connected, as in the diagram, with switch S open. The applied potential difference is $V_{ab} = +210$ V. (a) What is the potential difference V_{cd} ? (b) What is the potential difference across each capacitor after switch S is closed? (c) How much charge flowed through the switch when it was closed?

Figure P24.60



$$Q_T = V_{ab} C_{ab}$$

$$= 210 \times 4 = 840 \mu C$$

$$\left\{ \begin{array}{l} 3(6) = 2 \mu \\ 6(3) = \end{array} \right\}$$

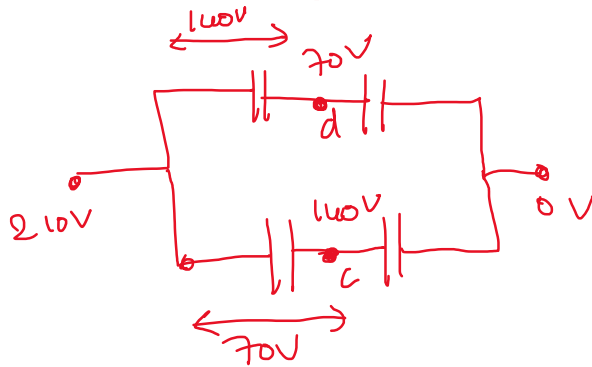
$$V_{ad} = \frac{42}{C_2} = \frac{420}{3} = 140 \text{ V}$$

420 μC

6 + 3

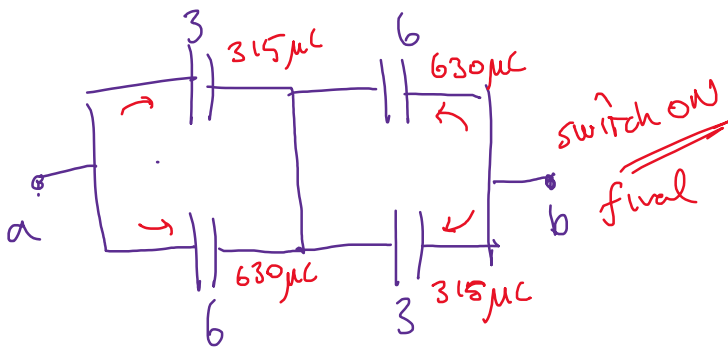
$$2 + 2 = 4 \mu$$

$$V_{ac} = \frac{Q_1}{C_1} = \frac{420}{6} = 70 \text{ V}$$

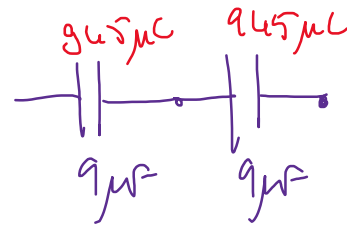


$$V_{cd} = V_d - V_c = 70 - 140 =$$

$$V_{cd} = V_c - V_d = +70 \text{ V}$$



switch on
final $\Rightarrow$

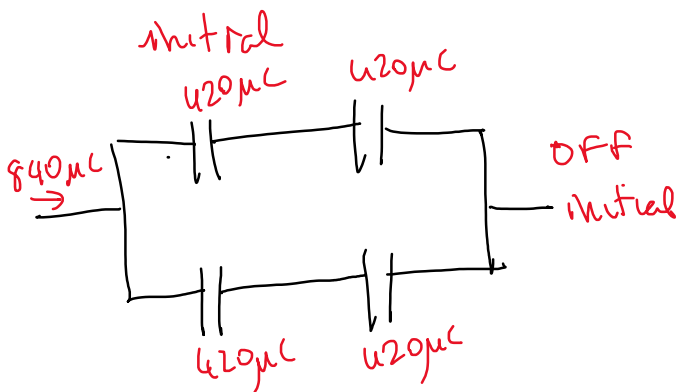


$$C_{eq} = 4.$$

$$V_{ab} = 210 \text{ V}$$

$$Q = 210(4.5) = 945$$

$$630 \mu\text{C} = \frac{945}{3} \cdot 2 \Rightarrow 6 \mu\text{F}$$



End of chapter

